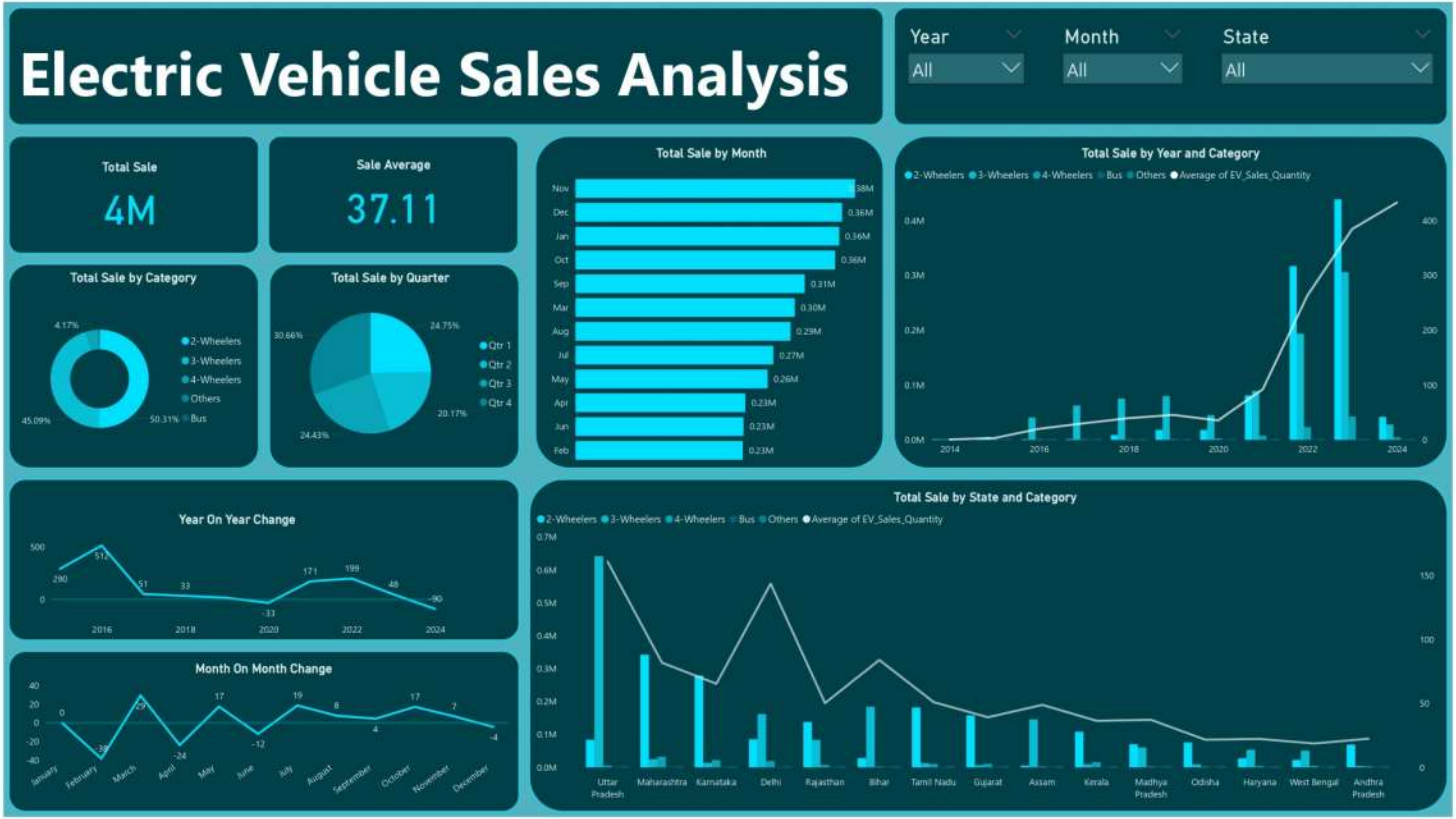
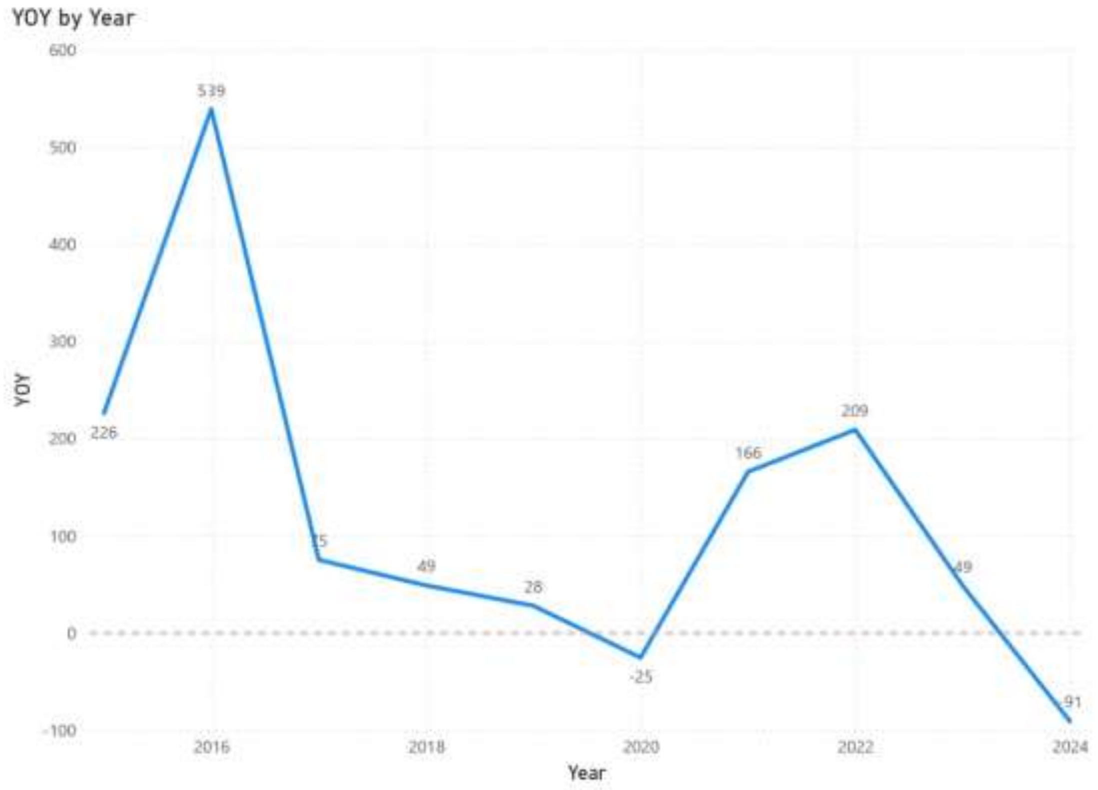
Electrical Vehicle Sales Analysis

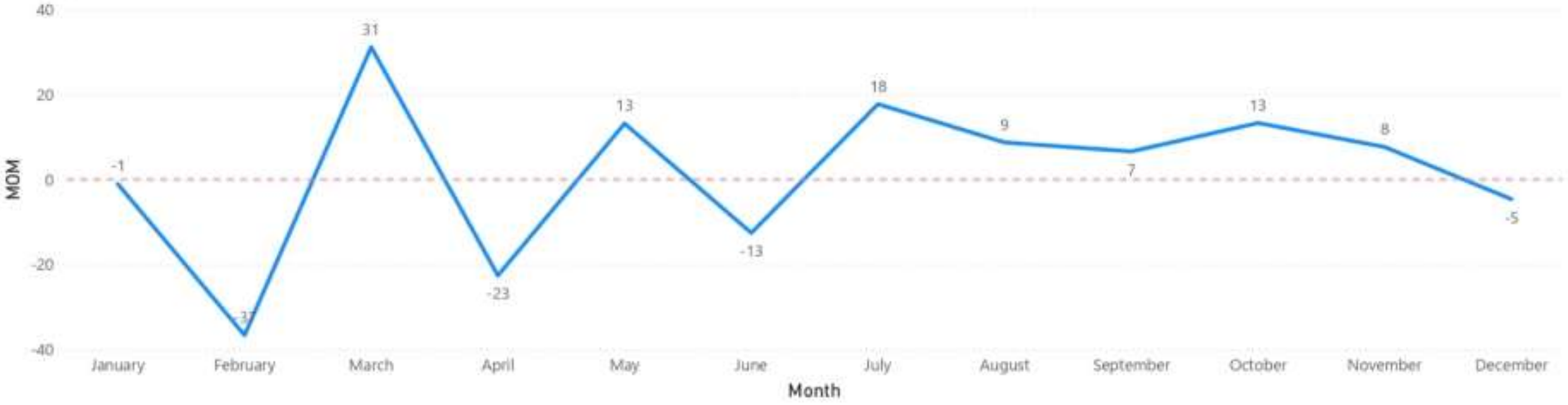

Electric Vehicle Sales Analysis
Year
All
Month
All
State
All
Total Sale
4M
Sale Average
37.11
Total Sale by Category
4.17%
45.09%
50.31%
2-Wheelers
3-Wheelers
4-Wheelers
Others
Bus
Total Sale by Quarter
30.66%
24.75%
20.17%
24.43%
Qtr 1
Qtr 2
Qtr 3
Qtr 4
Total Sale by Month
Nov 0.38M
Dec 0.36M
Jan 0.36M
Oct 0.36M
Sep 0.31M
Mar 0.30M
Aug 0.29M
Jul 0.27M
May 0.26M
Apr 0.23M
Jun 0.23M
Feb 0.23M
Total Sale by Year and Category
2-Wheelers 3-Wheelers 4-Wheelers Bus Others Average of EV_Sales_Quantity
0.0M 0.1M 0.2M 0.3M 0.4M
0 100 200 300 400
2014 2016 2018 2020 2022 2024
Year On Year Change
500
0
290 512 51 33 -33 171 199 48 -90
2016 2018 2020 2022 2024
Month On Month Change
40 20 0 -20 -40
0 -38 29 -24 17 -12 19 8 4 17 7 -4
January February March April May June July August September October November December
Total Sale by State and Category
2-Wheelers 3-Wheelers 4-Wheelers Bus Others Average of EV_Sales_Quantity
0.0M 0.1M 0.2M 0.3M 0.4M 0.5M 0.6M 0.7M
0 50 100 150
Uttar Pradesh
Maharashtra
Karnataka
Delhi
Rajasthan
Bihar
Tamil Nadu
Gujarat
Assam
Kerala
Madhya Pradesh
Odisha
Haryana
West Bengal
Andhra Pradesh

Year	Total_Sale	Last_Year_Sale	YOY
2014	2392		
2015	7805	2392	226.30
2016	49855	7805	538.76
2017	87420	49855	75.35
2018	130254	87420	49.00
2019	166819	130254	28.07
2020	124684	166819	-25.26
2021	331498	124684	165.87
2022	1024723	331498	209.12
2023	1525179	1024723	48.84
2024	143182	1525179	-90.61
Total	**3593811**	**3450629**	**4.15**

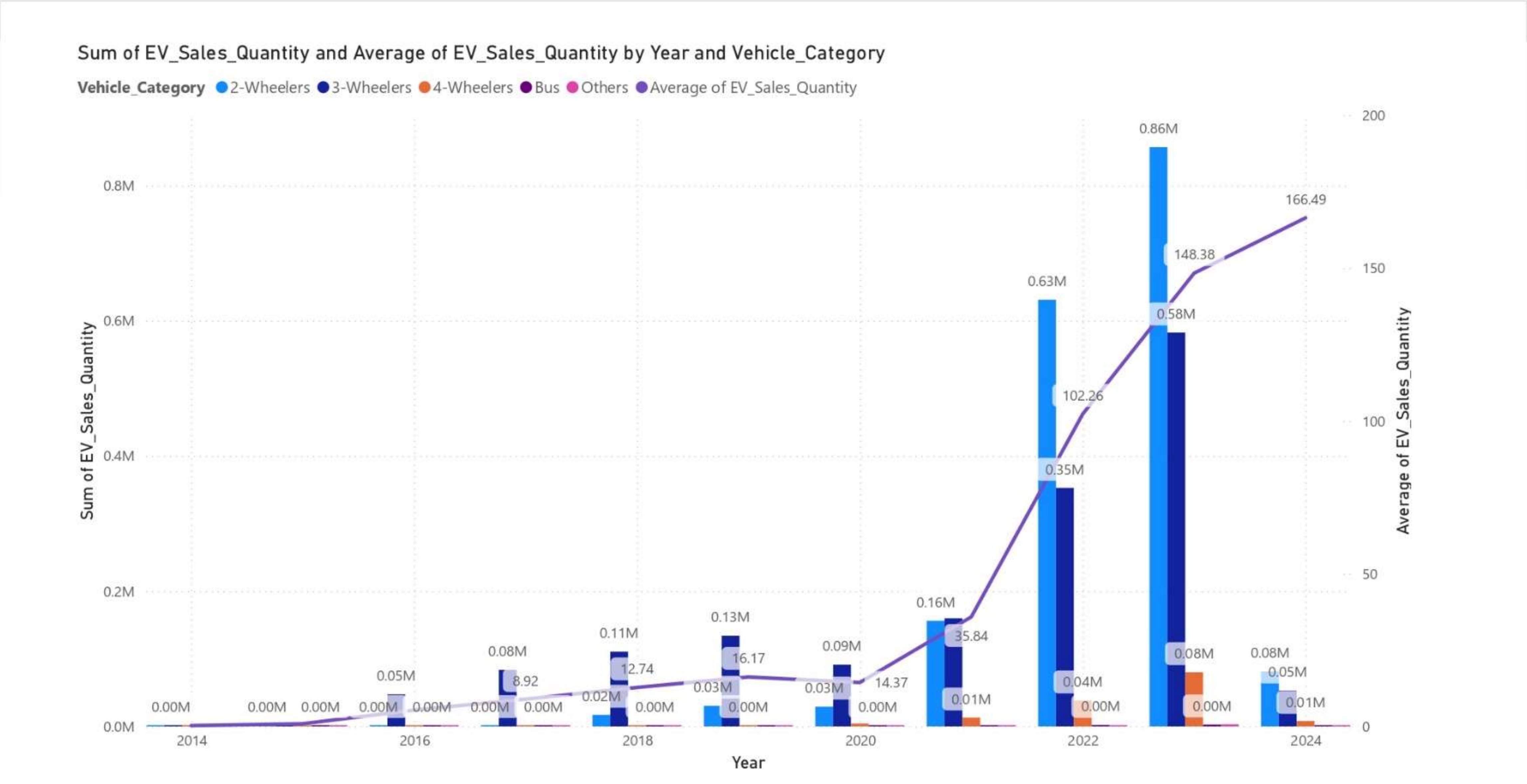


Month	Total_Sale	Last_Month_Sale	MOM
January	360703	364558	-1.06
February	228739	360703	-36.59
March	299888	228739	31.10
April	232194	299888	-22.57
May	262747	232194	13.16
June	229754	262747	-12.56
July	270473	229754	17.72
August	294022	270473	8.71
September	313433	294022	6.60
October	355083	313433	13.29
November	382217	355083	7.64
December	364558	382217	-4.62
Total	**3593811**	**3593811**	**0.00**

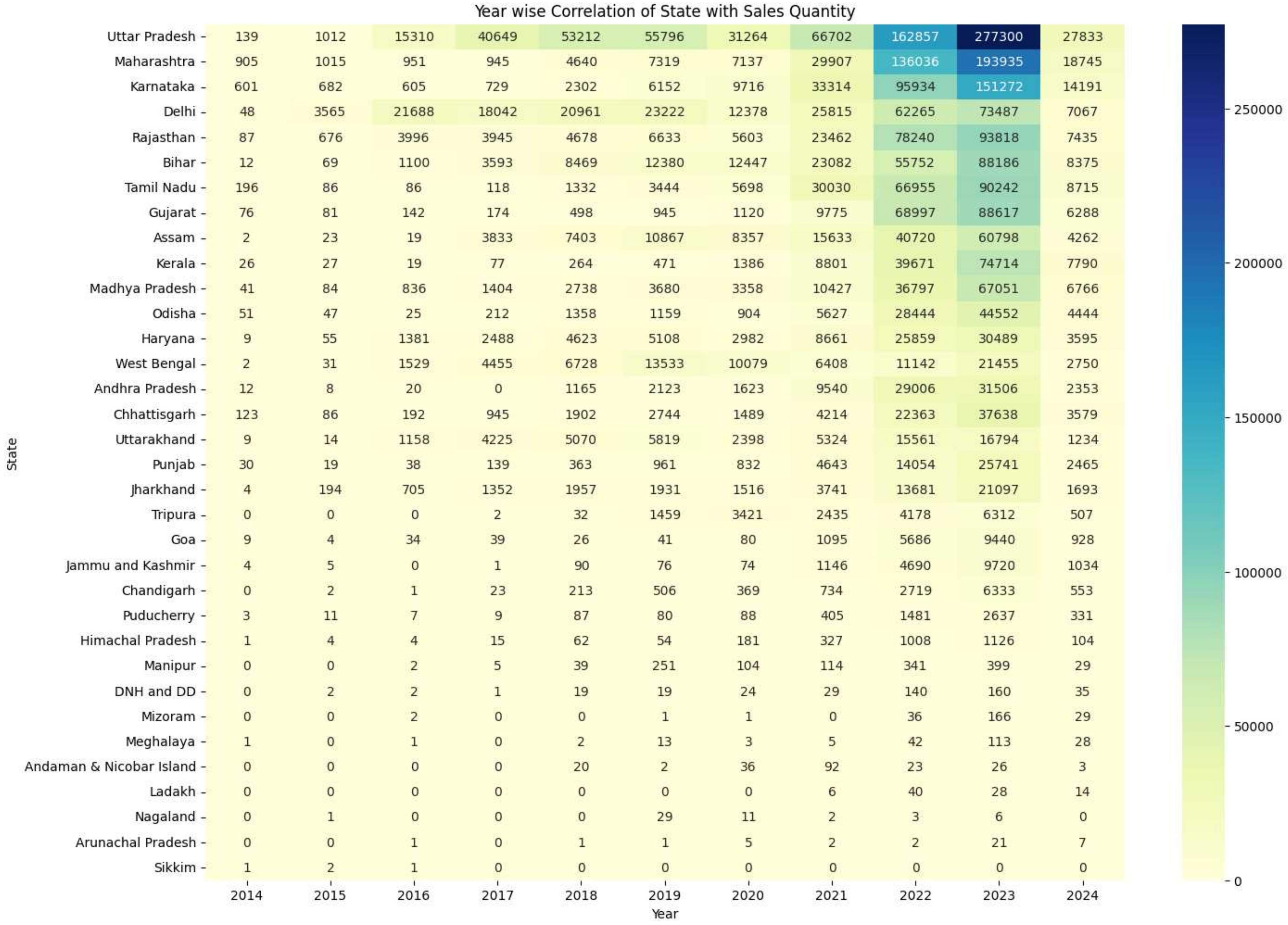
MOM by Month



- **EV Adoption is Rising:** The yearly sales trend shows strong growth in EV adoption across states, proving increasing public and business interest in EVs.
- **State-Wise Leaders:** Some states consistently dominate EV sales (e.g., U.P., Delhi, Maharashtra, Karnataka).
- **Vehicle Category Preference:** 2-Wheelers and 3-Wheelers form the majority of sales, while 4-Wheelers and Buses are still growing.
- **Seasonality & Demand Gaps:** Monthly/quarterly plots reveal spikes in sales during some months.



Sum of EV_Sales_Quantity and Average of EV_Sales_Quantity by Year and Vehicle_Category
Vehicle_Category 2-Wheelers 3-Wheelers 4-Wheelers Bus Others Average of EV_Sales_Quantity
Sum of EV_Sales_Quantity
Average of EV_Sales_Quantity
Year
0.0M
0.2M
0.4M
0.6M
0.8M
0
50
100
150
200
2014
2016
2018
2020
2022
2024
0.86M
0.63M
0.58M
0.35M
0.16M
0.13M
0.11M
0.09M
0.08M
0.05M
0.04M
0.03M
0.02M
0.01M
0.00M
166.49
148.38
102.26
35.84
16.17
14.37
12.74
8.92


Year wise Correlation of State with Sales Quantity
State
Year
2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024
Uttar Pradesh 139 1012 15310 40649 53212 55796 31264 66702 162857 277300 27833
Maharashtra 905 1015 951 945 4640 7319 7137 29907 136036 193935 18745
Karnataka 601 682 605 729 2302 6152 9716 33314 95934 151272 14191
Delhi 48 3565 21688 18042 20961 23222 12378 25815 62265 73487 7067
Rajasthan 87 676 3996 3945 4678 6633 5603 23462 78240 93818 7435
Bihar 12 69 1100 3593 8469 12380 12447 23082 55752 88186 8375
Tamil Nadu 196 86 86 118 1332 3444 5698 30030 66955 90242 8715
Gujarat 76 81 142 174 498 945 1120 9775 68997 88617 6288
Assam 2 23 19 3833 7403 10867 8357 15633 40720 60798 4262
Kerala 26 27 19 77 264 471 1386 8801 39671 74714 7790
Madhya Pradesh 41 84 836 1404 2738 3680 3358 10427 36797 67051 6766
Odisha 51 47 25 212 1358 1159 904 5627 28444 44552 4444
Haryana 9 55 1381 2488 4623 5108 2982 8661 25859 30489 3595
West Bengal 2 31 1529 4455 6728 13533 10079 6408 11142 21455 2750
Andhra Pradesh 12 8 20 0 1165 2123 1623 9540 29006 31506 2353
Chhattisgarh 123 86 192 945 1902 2744 1489 4214 22363 37638 3579
Uttarakhand 9 14 1158 4225 5070 5819 2398 5324 15561 16794 1234
Punjab 30 19 38 139 363 961 832 4643 14054 25741 2465
Jharkhand 4 194 705 1352 1957 1931 1516 3741 13681 21097 1693
Tripura 0 0 0 2 32 1459 3421 2435 4178 6312 507
Goa 9 4 34 39 26 41 80 1095 5686 9440 928
Jammu and Kashmir 4 5 0 1 90 76 74 1146 4690 9720 1034
Chandigarh 0 2 1 23 213 506 369 734 2719 6333 553
Puducherry 3 11 7 9 87 80 88 405 1481 2637 331
Himachal Pradesh 1 4 4 15 62 54 181 327 1008 1126 104
Manipur 0 0 2 5 39 251 104 114 341 399 29
DNH and DD 0 2 2 1 19 19 24 29 140 160 35
Mizoram 0 0 2 0 0 1 1 0 36 166 29
Meghalaya 1 0 1 0 2 13 3 5 42 113 28
Andaman & Nicobar Island 0 0 0 0 20 2 36 92 23 26 3
Ladakh 0 0 0 0 0 0 0 6 40 28 14
Nagaland 0 1 0 0 0 29 11 2 3 6 0
Arunachal Pradesh 0 0 1 0 1 1 5 2 2 21 7
Sikkim 1 2 1 0 0 0 0 0 0 0 0
250000
200000
150000
100000
50000
0

- **Statistical Testing Findings:**
- Hypothesis Testing:
- EV sales are not equal in every state → Some states show much higher adoption than others.
- A/B Testing:
- Sales trend changed after 2020 → Growth in EV adoption became much stronger.
- Chi-Square Testing:
- State and EV type are connected → Different states prefer different categories of EVs (like 2-wheelers, 3-wheelers, or 4-wheelers).

- **Machine Learning Insights:**
- The Decision Tree model was able to predict Vehicle Category with good accuracy using features like Year, State and Sales. This means sales patterns are predictable, which is valuable for demand forecasting.
- **Policy & Business Implication:**
- States with low adoption can be targeted with awareness programs, incentives, and better charging networks.
- EV manufacturers should focus more on 2W/3W markets for volume and 4W/Buses for long-term growth.
- **Future Growth Opportunity:**
- Combining trends, testing, and ML shows that EV sales will continue to grow, but growth will not be uniform. Businesses, governments, and investors must plan state-specific and category-specific strategies instead of a one-size-fits-all approach.